TRACE-Q: Supplementary Material

1 Paper-Level Results

Table S1: Per-paper counts derived from the final human workbooks.

Year	Paper DOI suffix	Candidates	Retained	Primary	Strict	Secondary
2018	jama.2017.21906	7	6	1	0	0
2018	jama.2018.0156	8	7	1	1	0
2018	jama.2018.0948	12	11	0	0	0
2018	jama.2018.14280	10	10	1	1	0
2018	jama.2018.14282	16	16	3	3	1
2018	jama.2018.17075	15	15	2	1	0
2018	jama.2018.18020	22	21	5	4	0
2018	jama.2018.6496	13	11	0	0	0
2018	jama.2018.8802	5	4	1	1	0
2018	jama.2018.9128	7	6	0	0	0
2019	jama.2019.0556	11	11	1	0	0
2019	jama.2019.10517	5	5	0	0	0
2019	jama.2019.12618	10	10	0	0	0
2019	jama.2019.14630	9	9	1	1	0
2019	jama.2019.14901	5	5	0	0	0
2019	jama.2019.17380	18	18	0	0	0
2019	jama.2019.2210	7	5	0	0	0
2019	jama.2019.4755	1	1	0	0	0
2019	jama.2019.7505	13	12	4	4	0
2024	jama.2024.0318	14	14	0	0	0
2024	jama.2024.0572	19	16	5	5	0
2024	jama.2024.11057	5	5	0	0	0
2024	jama.2024.12829	7	7	1	1	0
2024	jama.2024.19585	7	5	0	0	0
2024	jama.2024.2302	6	6	1	1	0
2024	jama.2024.23898	13	13	0	0	0
2024	jama.2024.25786	15	14	1	1	0
2024	jama.2024.4183	14	14	1	1	0
2024	jama.2024.6063	13	13	2	2	0
2025	jama.2024.24764	5	5	1	1	0
2025	jama.2024.27643	16	13	2	2	1
2025	jama.2025.11178	18	17	3	3	0
2025	jama.2025.15185	6	6	1	1	0
2025	jama.2025.15440	1	1	0	0	0
2025	jama.2025.16450	6	5	0	0	0
2025	jama.2025.19563	13	10	0	0	0
2025	jama.2025.19843	7	7	0	0	0
2025	jama.2025.20765	19	19	1	1	1
2025	jama.2025.24175	14	14	2	2	1
2025	jama.2025.4390	13	13	0	0	0
2025	jama.2025.7583	4	4	0	0	0
2025	jama.2025.7710	3	3	0	0	0
2025	jama.2025.9110	3	2	0	0	0
2025	jama.2025.9663	5	4	0	0	0

DOI prefix is 10.1001/. Primary = substantive confirmed/probable statistical issues; strict = substantive confirmed statistical issues; secondary = substantive report-internal inconsistencies. All three require an original report-only relationship.

Table S2: Candidate dispositions using the final human review domains.

Review domain	Candidates	Retained	Excluded	Retained (%)
Statistical results	173	155	18	89.6
Report internal inconsistency	12	12	0	100.0
Protocol–report	216	214	2	99.1
Planning material	39	32	7	82.1

Five candidates changed domain during human review. Protocol–report differences and planning-material findings are descriptive; they do not enter the temporal endpoints.

2 Adjudication and Analysis Details

The analysis reads the four completed Excel workbooks directly, using stable candidate keys to join original relationship codes. It checks unique keys, row totals, completion, and required domain-specific judgments. All 440 supplied human decisions are preserved. Final human domains determine descriptive summaries; original document relationships additionally constrain temporal eligibility. Candidate `2024::jama.2024.11057::C003`, for example, was retained as a substantive report-internal inconsistency after domain editing, but its original `P_R` relationship excludes it from the secondary temporal endpoint. It remains in the descriptive human-domain total.

The 149 confirmed report-material issues of any substantiveness occurred in 38 of 44 papers. This broader count includes minor presentation issues and differs from the principal endpoint of 41 confirmed or probable substantive statistical issues. The broad count is descriptive and is not a detection-accuracy measure.

2.1 Illustrative Human Decisions

For `jama.2018.14280`, C001 concerned the identity of a reported binary risk ratio and was retained as a confirmed substantive statistical issue with direct potential impact on extraction. For `jama.2017.21906`, C001 compared a planned medication composite containing statin use with the reported composite; the report disclosed that statin data were not collected. This was retained as a substantive protocol–report difference without an inference of unjustified change. For `jama.2018.0156`, C002 was excluded because secondary-outcome and safety-analysis descriptions could coexist within the supplied plans. These examples summarize final adjudications, rather than proposing corrected results.

2.2 Full Period Comparisons

Table S3 gives the complete set of exploratory comparisons, including binary sensitivity endpoints omitted from the main table for space. All tests reallocate papers between groups of 19 and 25; candidates remain grouped within their papers. Exact probabilities count all allocations with a statistic at least as extreme in absolute value, including ties. Bootstrap intervals resample papers within periods and are not permutation intervals.

2.3 Interpretation of Review Fields

3 Reproducible Computation

The files `candidate_results.csv` and `paper_results.csv` preserve analysis variables. The file `period_comparisons.csv` contains unrounded estimates, intervals, and exact permutation probabilities. Original candidate domain and relationship codes are stored separately from final human labels. The R script cross-checks its dynamic-programming calculation against direct enumeration of a small example with tied and

Table S3: Exploratory comparison of report-material outcomes between publication periods.

Outcome	2018–2019	2024–2025	Difference (95% CI)	*P*
Substantive statistical (count)	1.05	0.84	-0.21 (-1.02, 0.56)	0.654
Substantive statistical (any)	10/19	12/25	-4.6 (-33.7, 25.7)	1.000
Substantive report inconsistency (count)	0.05	0.12	0.07 (-0.09, 0.24)	0.622
Substantive report inconsistency (any)	1/19	3/25	6.7 (-9.1, 24.0)	0.622
Confirmed statistical only (count)	0.84	0.84	0.00 (-0.77, 0.74)	1.000
Confirmed statistical only (any)	8/19	12/25	5.9 (-24.4, 35.2)	0.766
Combined substantive report (count)	1.11	0.96	-0.15 (-1.02, 0.68)	0.754
Combined substantive report (any)	10/19	12/25	-4.6 (-33.7, 25.7)	1.000

Count rows show the mean number per paper; any rows show papers with at least one eligible issue. Differences are later minus earlier, in issues per paper or percentage points, respectively. Principal-endpoint eligibility includes confirmed or probable statistical issues judged substantive. The confirmed-only and combined endpoints are sensitivity analyses. Confidence intervals use 20,000 paper bootstrap resamples within periods; two-sided permutation *P* values are exact and unadjusted. No protocol–report or planning-material finding contributes.

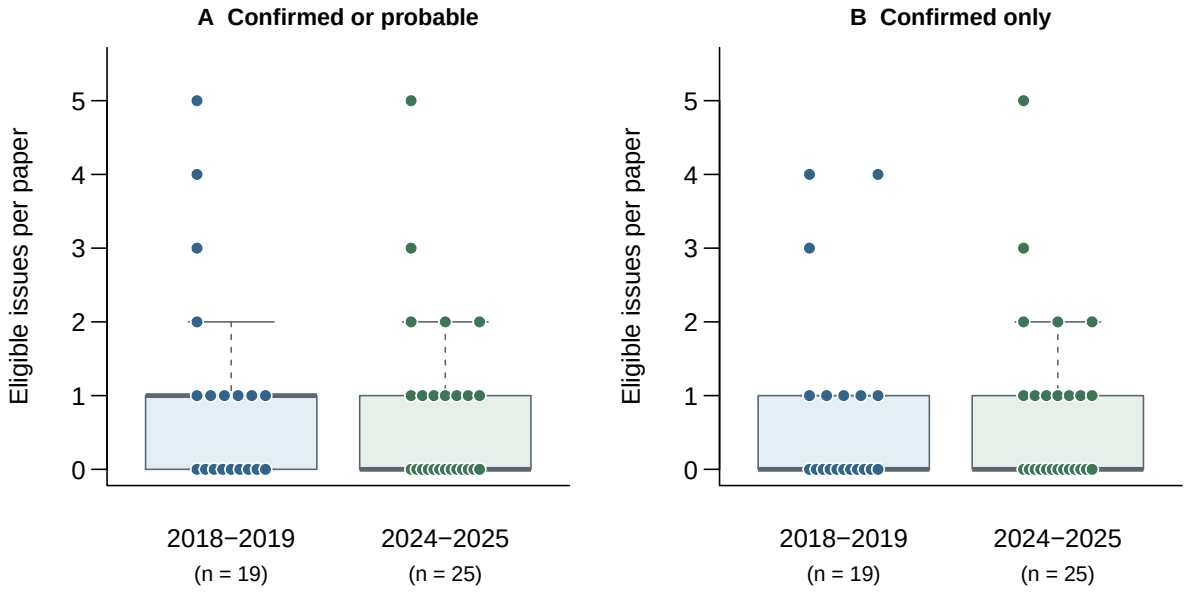

Figure S1: Substantive statistical issues by publication period. Each point represents a paper, including zero counts. Panel A includes confirmed and probable issues; Panel B includes confirmed issues only. Boxes show medians and interquartile ranges; whiskers extend to observations within 1.5 interquartile ranges.

Table S4: Operational interpretation of final human review fields.

Field	Interpretation used in the analysis
Disposition	Retain, exclude, or discuss; blank means unanswered. Retention alone is not confirmation.
Statistical issue	Confirmed, probable, not an issue, or indeterminate; required for retained statistical candidates.
Substantiveness	Substantive, minor presentation, or indeterminate; separate from evidential certainty.
Downstream impact	Optional direct, possible, unlikely, or unknown impact; not an eligibility criterion or evidence of actual harm.
Internal inconsistency	Substantive, minor presentation, no inconsistency, or indeterminate; used for report-internal and planning-material candidates.
Protocol interpretation	Optional description of a difference or ambiguity; excluded from temporal endpoints regardless of apparent importance.
Completion	`primary_done` indicates a completed review entry, not scientific confirmation.

zero counts; an independent implementation reproduced all eight reported permutation probabilities. All 44 papers enter the analysis, including those with no eligible finding. The manuscript tables use final workbooks rather than historical preliminary counts in project correspondence.

The source archive contains 188 PDFs (5,559 pages) and one Excel worksheet. Thus, 5,560 is the number of mixed source units, not the number of PDF pages. Source-file and report hashes are retained with the analysis. The project notes document the initial feasibility exercise and the rationale for separating broad descriptive review from narrower temporal outcomes; the main paper does not treat these notes as published evidence or the pilot as independent validation.